SIMATS ENGINEERING

Engineer to Excel

# TECH STAR SUMMIT 2024

Name: Ms. D. Pavithra
Register Number: 192210339
Guided by Dr. R. Gnajayeyaraman

# An Efficient Analysis For Predicting Kyphosis Disease Using XgBoost Algorithm In Comparison With KNN Algorithm To Improve Accuracy

## INTRODUCTION

- To Compare XGBoost and KNN algorithms for Kyphosis disease prediction, aiming to identify the more accurate predictor.
- It is Crucial for medical diagnostics, this study informs healthcare decisions, potentially improving patient outcomes by optimizing diagnostic accuracy.
- The research aids in medical diagnosis, specifically in Kyphosis disease detection and treatment planning, optimizing diagnostic processes for timely interventions.
- XGBoost is known for scalability and accuracy, it employs decision trees for robust predictions and KNN utilizes the principle of similarity for classification, making it simple yet effective in predictive tasks.
- The kyphosis detection dataset utilized is sourced from Kaggle, a reputable platform, Enabling seamless data sharing and supporting data driven investigation.

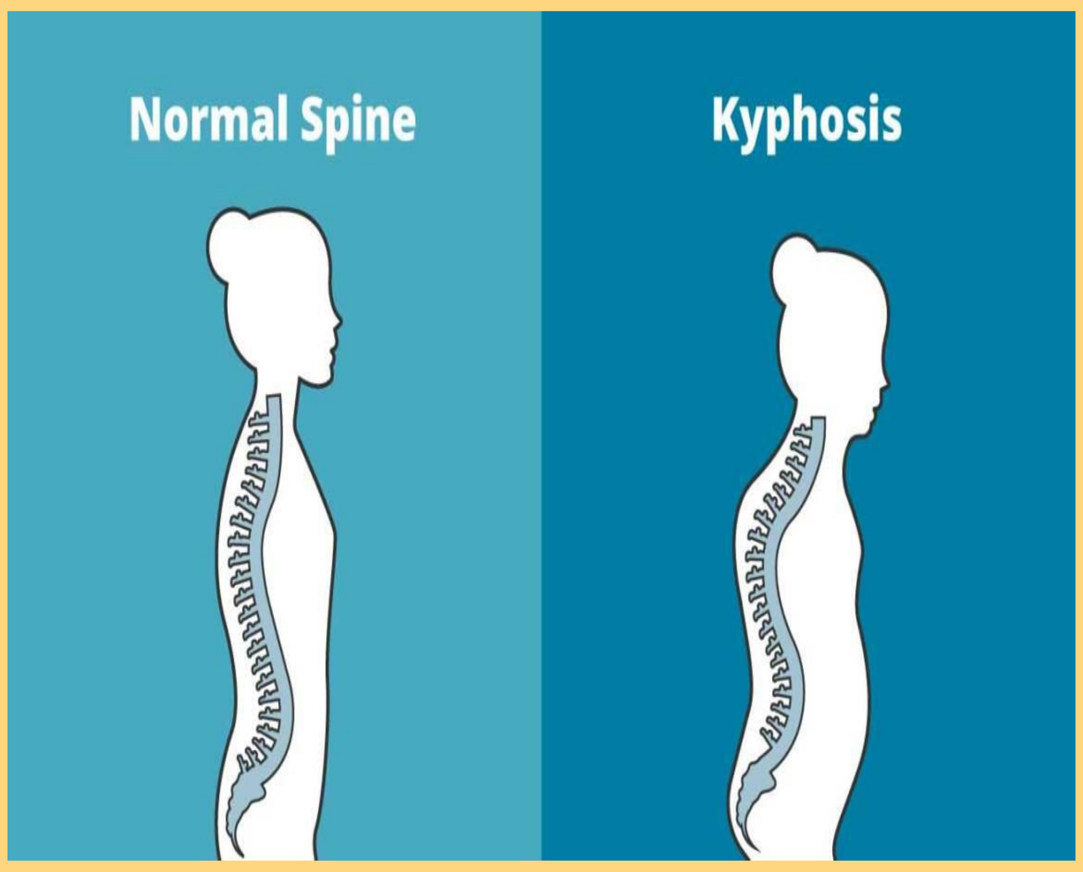
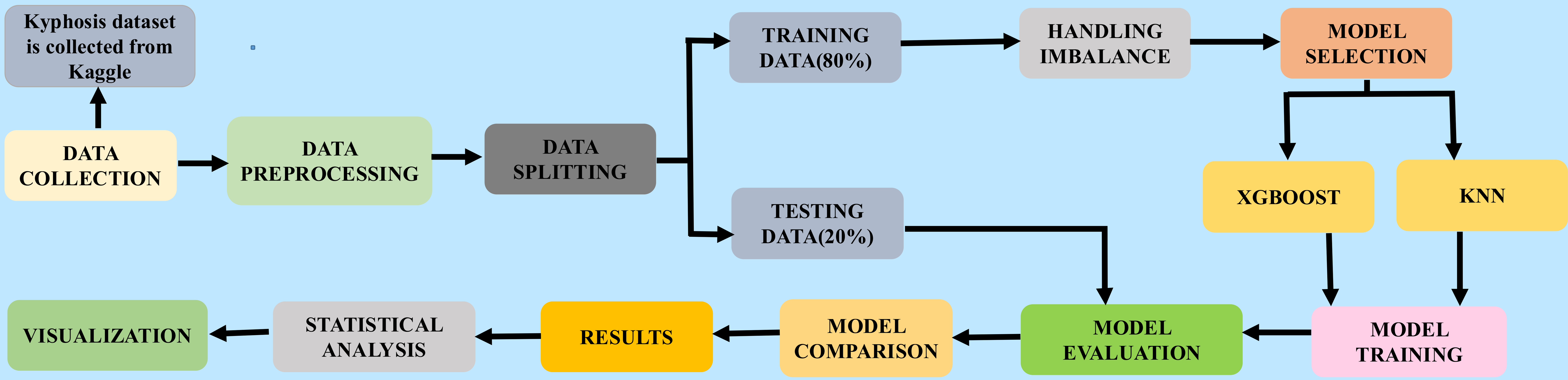



Fig. 1:Normal and affected spine by kyphosis.

## MATERIALS AND METHODS




## RESULTS

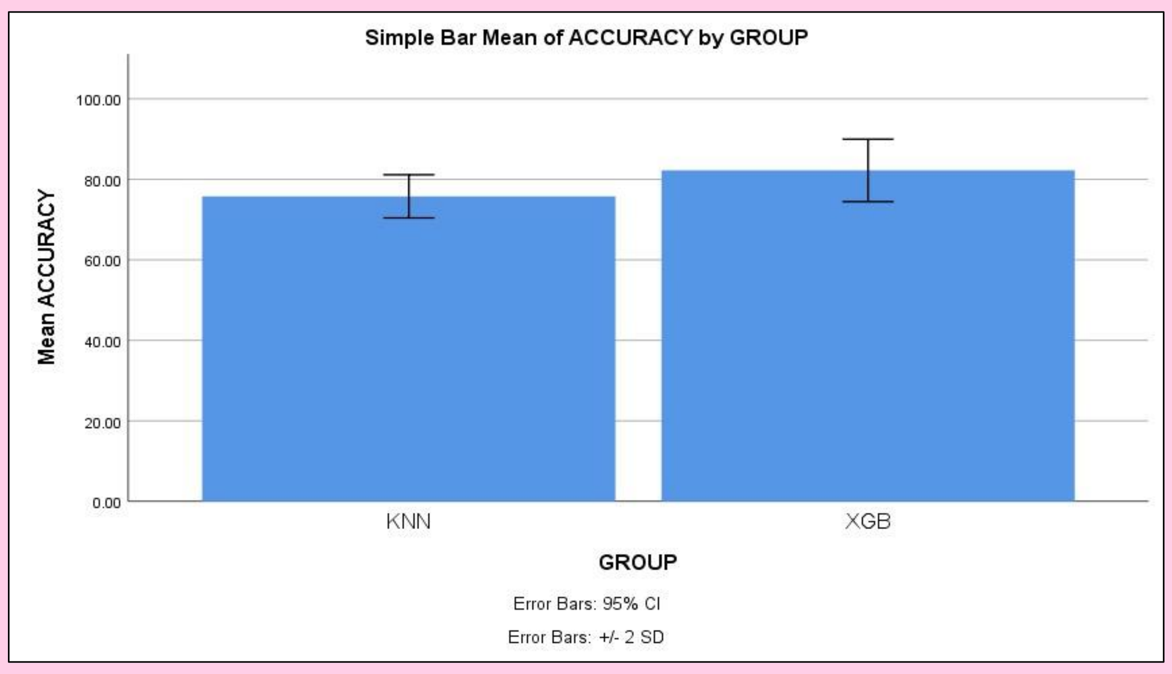



Fig. 2:Comparison of kyphosis detection between Xgboost and KNN.

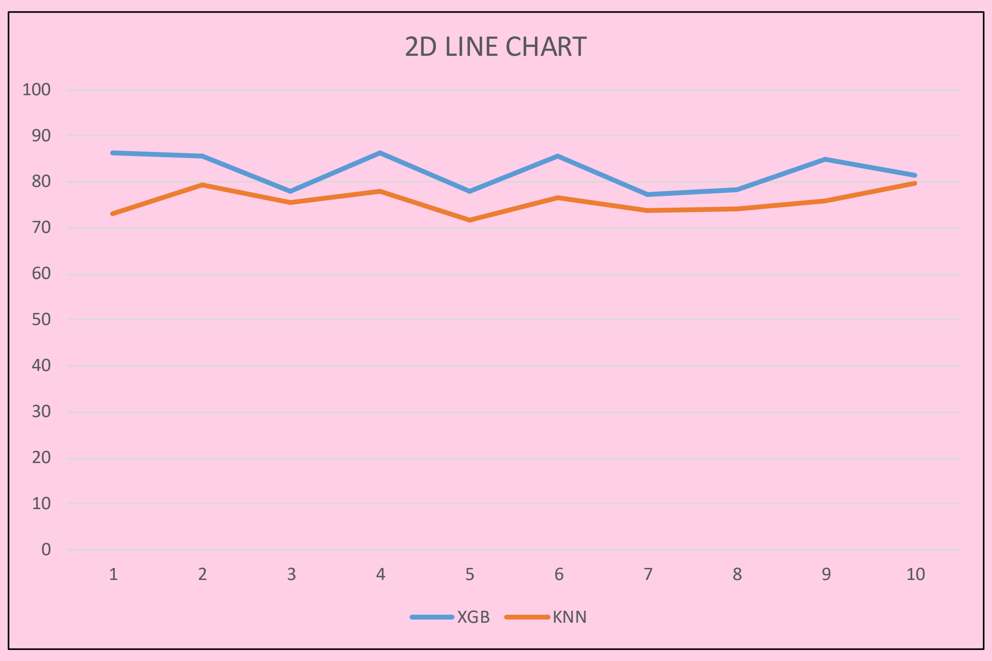



Fig. 3: Comparison of accuracy values of kyphosis detection with Xgboost and KNN.

| GROUP | | N | Mean | Std. Deviation | Std. Error Mean |
|---|---|---|---|---|---|
| ACCURACY | XGB | 10 | 82.2140 | 3.87330 | 1.22484 |
| | KNN | 10 | 75.7720 | 2.68483 | 0.84902 |

Table 1: Comparison of the mean accuracy of kyphosis detection with Xgboost and KNN.

## DISCUSSION AND CONCLUSION

- Based on T-test statistical analysis, the significance value of p=0.001(independent sample T-test p<0.05) is obtained and shows that there is a statistical significant difference between the group 1 and group 2(group 1-Xgboost & group 2-KNN).
- The KNN exhibited an accuracy rate of 76.47%,whereas the XgBoost algorithm, demonstrated an accuracy of 82.35% in predicting kyphosis disease in individuals.
- Further research could focus on optimizing XGBoost's computational efficiency and enhancing its interpretability while improving predictive performance in kyphosis detection
- Age, gender, and other factors can affect the accuracy of kyphosis prediction using machine learning. However, limitations arise from data availability, quality, potential bias, and the need for validation on diverse populations.
- This study decisively demonstrates XGBoost's superiority over KNN in Kyphosis disease prediction, highlighting the pivotal role of algorithm selection in refining medical diagnostics and optimizing patient care pathways.

## BIBLIOGRAPHY

- Dankwa, Stephen, and Wenfeng Zheng. 2019. "Special Issue on Using Machine Learning Algorithms in the Prediction of Kyphosis Disease: A Comparative Study" *Applied Sciences* 9, no. 16: 3322. https://doi.org/10.3390/app9163322.
- Lee CH, Jo DJ, Oh JK, Hyun SJ, Park JH, Kim KH, Bae JS, Moon BJ, Lee CK, Shin MH, Jang HJ, Han MS, Kim CH, Chung CK, Moon SM. Development and Validation of an Online Calculator to Predict Proximal Junctional Kyphosis After Adult Spinal Deformity Surgery Using Machine Learning. Neurospine. 2023 Dec;20(4):1272-1280. doi: 10.14245/ns.2342434.217. Epub 2023 Dec 31. PMID: 38171294; PMCID: PMC10762414.
- Chauhan, A.S.; Lilhore, U.K.; Gupta, A.K.; Manoharan, P.; Garg, R.R.; Hajjej, F.; Keshta, I.; Raahemifar, K. Comparative Analysis of Supervised Machine and Deep Learning Algorithms for Kyphosis Disease Detection. Appl. Sci. 2023, 13, 5012.https://doi.org/10.3390/app13085012.
- Hussein, Yehia Y., and Muhammad Mohsin Khan. "Using Artificial Intelligence to Predict the Development of Kyphosis Disease: A Systematic Review." Cureus 15, no. 11 (2023).doi:10.7759/cureus.48341.